Earthquake Insurance Analysis Project — California (2023)

UNDERINSURED EARTHQUAKE RISK — CALIFORNIA COUNTIES

Geospatial & Insurance Modeling to Identify

High-Risk, Underinsured Regions

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Project Overview

This project analyzes earthquake insurance underinsurance across 58 California counties by combining seismic history, fault proximity, housing vulnerability, demographics, and insurance data. Using feature engineering and an XGBoost predictive model, the analysis estimates county-level take-up gaps and identifies regions with the highest financial exposure.

Key Takeaways

- Built a unified geospatial dataset from **USGS, Census ACS, CDI,** and **FEMA** sources
- Created engineered features capturing **seismic risk, housing age, vulnerability,** and **insurance behavior**
- Developed an XGBoost model achieving **$R^2 \approx 0.53$** with strong alignment between predicted and actual take-up
- Identified **16 high-impact counties** with significant underinsurance exposure
- **Estimated $64.2M** in underinsured cost statewide
- Designed an interactive Tableau dashboard for decision-making and communication

Why Earthquake Underinsurance Matters

California faces frequent seismic activity, yet earthquake insurance adoption remains low and uneven across the state. Many communities located near active faults carry high physical risk but lack sufficient financial protection — resulting in large post-disaster losses and increased reliance on public assistance.

Core Challenges

- Significant gap between **earthquake hazard** and **insurance coverage**
- High-risk and vulnerable counties may struggle to recover after major events
- Insurers, policymakers, and emergency managers need granular visibility into where financial exposure is highest
- Traditional hazard-based assessments don't capture the full picture — county-level decisions require deeper, data-driven modeling

Project Goal

Objective Statement

Develop a data-driven, county-level framework to measure earthquake underinsurance, estimate financial exposure, and pinpoint regions where seismic risk and low insurance take-up create the greatest vulnerability.

Specific Goals

- Integrate **seismic, demographic, housing**, and **insurance** datasets into a unified geospatial framework
- Quantify how **fault exposure, seismic history, housing age, socioeconomic factors**, and **insurance behavior** influence take-up
- Build a **predictive model** to estimate **expected take-up** for each county
- Identify counties where **actual take-up** falls below **modeled expectations**
- Translate model results into **underinsurance tiers** and **estimated dollar exposure**
- Deliver insights through an **interactive Tableau dashboard**

Data Sources & Integrations

To build a comprehensive view of earthquake underinsurance across California,
I integrated multiple authoritative datasets into a unified geospatial framework.

Primary Data Sources

- **USGS Earthquake Catalog (2023)**: historical seismic events, magnitude, depth, and frequency
- **USGS Quaternary Faults Database**: fault geometry, proximity, and total fault length
- **California Department of Insurance (CDI)**: earthquake policy counts and premiums by county
- **U.S. Census Bureau ACS 5-Year Estimates (2023)**: demographics, poverty, race/ethnicity, housing age, tenure, vacancy
- **FEMA Public Assistance Data**: federal recovery assistance as a proxy for hazard burden

Processing Steps

- Spatial joins and geospatial alignment using **GeoPandas**
- Normalization and harmonization of ACS, CDI, and FEMA variables
- Feature engineering for **seismic**, **socioeconomic**, and **insurance behavior** metrics

Data Preparation & Feature Engineering

After merging all datasets, I engineered features capturing the key drivers of seismic exposure, housing vulnerability, socioeconomic conditions, and insurance behavior across California counties.

Engineered Feature Groups

SEISMIC EXPOSURE

- Earthquake count & max magnitude
Fault proximity (nearest km)
- Centroid-to-fault distance
Fault intersections (binary)
- Total fault length (km)
Fault orientation (azimuth)

HOUSING VULNERABILITY

- % Pre-1980 housing
- % 1980–1999 housing
- % 2000+ housing
- Owner vs. renter occupancy
- Median home value
- Vacancy types (%): for-rent, for-sale, seasonal

SOCIOECONOMIC

- Total population
- Median income
- Poverty rate (%)
- Age structure (% 0–17, 18–64, 65+)
- Migration (%)
- ACS population composition (non-sensitive)

INSURANCE BEHAVIOR

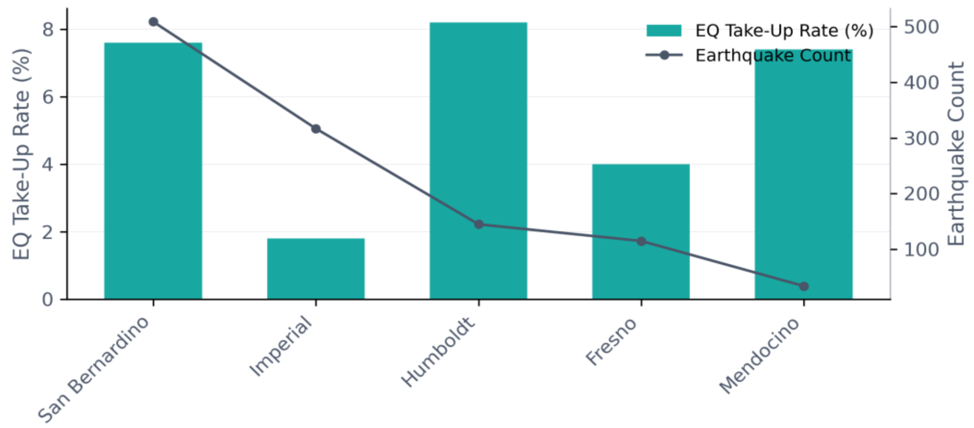
- Earthquake policy counts
- Earthquake take-up rate (%)
- Homeowners insurance coverage (%)
- Total homeowners policies
- CEA policies (new & renewed)
- FAIR Plan policies
- CDI exposure metrics
  - Direct liability (2018)
  - Direct insured PML (2018)
  - Modeled loss ratio (%)

Outcome: Engineered dataset (gdf_ca_extended_v3) for EDA, composite scoring, and modeling.

Exploratory Data Analysis (EDA)

Key Findings

- **Normalization was essential:** County size differences skewed many raw variables (population, housing counts, FEMA funding, pre-1980 housing), making percentage-based features more comparable.
- **Seismic exposure influences take-up, but inconsistently:** Counties near faults show higher take-up, yet several high-fault counties remain underinsured.
- **Housing age is a strong signal:** Older housing correlates positively with earthquake insurance take-up; newer homes correlate negatively, reflecting lower perceived vulnerability.
- **Socioeconomic conditions matter more than seismic risk:** Poverty, vacancy types, and age structure correlate more strongly with take-up.
- **Vacancy composition is highly predictive:** Seasonal housing correlates with higher take-up; migrant and “other vacant” categories correlate negatively.
- **Spatial EDA revealed clear clusters:** Maps and heatmaps highlighted counties with high seismic exposure but low insurance participation (e.g., San Bernardino, Imperial, Humboldt, Mendocino).



Result: A deep understanding of which variables meaningfully relate to earthquake insurance take-up — guiding **variable selection**, **composite score design**, and the **feature set** for predictive modeling.

Predictive Modeling Approach

The goal of modeling was to estimate the expected earthquake insurance take-up rate for each county using engineered demographic, housing, seismic, and insurance features.

1. Feature Selection

- Evaluated multiple feature sets (20-feature baseline, 10-feature subset, correlation-pruned)
- Final model used **correlation-pruned features** to reduce multicollinearity and improve generalization

2. Model Choice

- Tested multiple regressors
- **XGBoost** with a preprocessing pipeline delivered the strongest performance
- Tuned using **GridSearchCV** (learning rate, depth, regularization)

3. Model Performance

- **Best model:** XGBoost (correlation-pruned features)
- **R^2 = 0.536**
- Strong relationship between predicted and actual take-up rates

4. Model Outputs

- Predicted take-up rate
- Take-up gap (predicted − actual)
- Underinsurance tiers
- Dollar exposure estimates

Predictive Modeling Results

Best Model

- **XGBoost Regressor** using a **correlation-pruned feature set**
- Performance: **R^2 = 0.536** (cross-validation)
- Strong linear alignment between predicted and actual take-up

Top Predictive Features

- **Insurance & financial:** homeowners_coverage_pct, median_income, modeled_loss_ratio_2018_pct
- **Seismic:** eq_max_mag, fault_length_km, centroid_to_fault_km
- **Socioeconomic & housing:** age_0_17_pct, below_poverty_pct, migrant_pct, pre_1980_pct, for_sale_pct, seasonal_pct

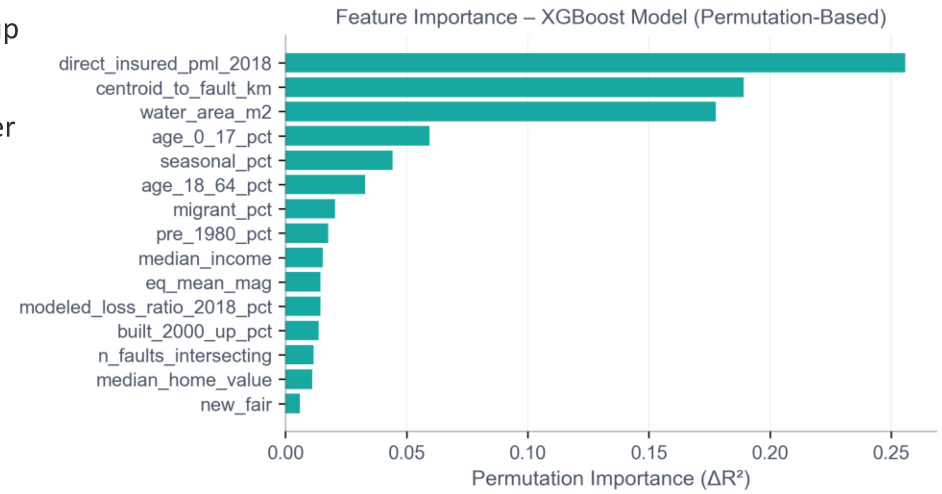
Residual Patterns

- Coastal counties: model **underestimates** take-up → higher-than-expected coverage
- Inland counties: model **overestimates** → deeper underinsurance or unmodeled barriers
- Underestimated: **Mono, Inyo, San Luis Obispo**
- Overestimated: **Imperial, Glenn, Tehama**

Underinsurance Gap

- Estimated statewide gap: **$60M+**
- **Tier 1 & 2 (16 counties)** ≈ **93%** of statewide underinsurance
- Examples:
  - **High Gap & High Cost:** Santa Clara, Alameda, Fresno, Merced, Imperial
  - **High Cost Only:** Los Angeles, San Bernardino, San Joaquin, Riverside

Model Diagnostics — Feature Impact



Outcome: A validated model pinpointing counties where expected coverage and actual behavior diverge — guiding targeted outreach, pricing strategy, and seismic mitigation.

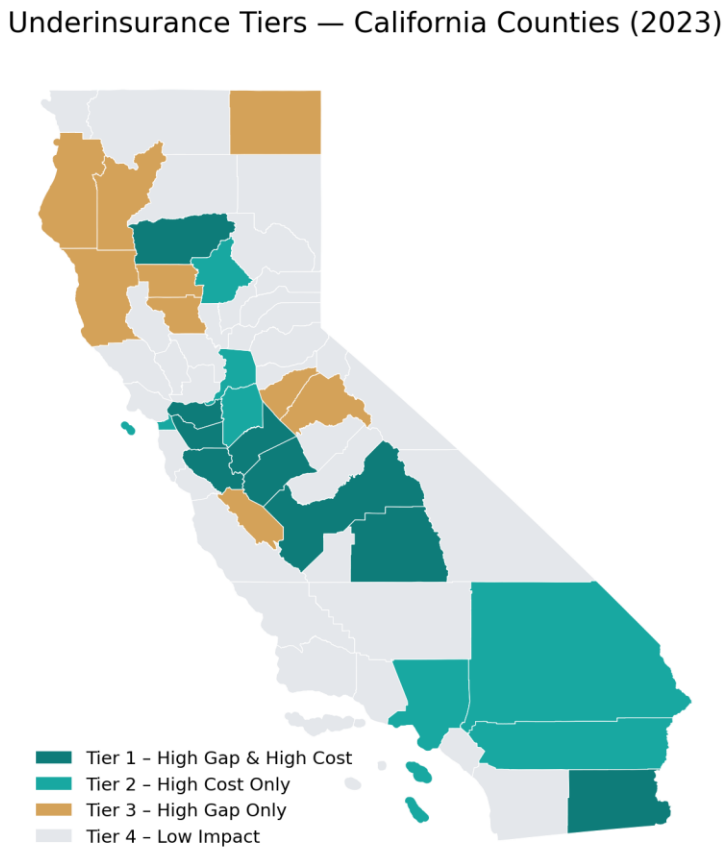
Underinsurance Tiers & Exposure Score

To turn model predictions into actionable insights, I built a two-dimensional classification combining:

1. Insurance Gap — predicted vs. actual take-up
2. Financial Exposure — calculated as underinsured households × ~$800 average policy cost.

Tier Definitions

- Tier 1 — **High Gap & High Cost** largest gaps + highest financial exposure
- Tier 2 — **High Cost Only** large exposure even with moderate gaps
- Tier 3 — **High Gap Only** underserved regions with smaller financial footprint
- Tier 4 — **Low Impact** lower risk and minimal exposure



Outcome
Identified **16 high-impact counties** (Tiers 1–2) that account for most of California's underinsurance exposure.

Key Findings from the Analysis

Statewide Insights

- California has an estimated **$64.2M** in underinsured earthquake exposure
- **Los Angeles, Santa Clara, and Contra Costa** have the highest financial exposure

Behavior vs. Risk Misalignment

- Many inland and rural counties show **high take-up gaps** despite moderate hazard levels → likely affordability and socioeconomic barriers
- The model reveals a strong mismatch between **expected risk-based behavior** and **actual insurance participation**

Key Predictors of Take-Up

- **Fault proximity** is predictive but **not sufficient alone**
- **Older housing (pre-1980)** strongly correlates with higher take-up
- **Vacant and rental-heavy markets** show lower take-up and greater underinsurance

Interactive Tableau Dashboard

To make the results accessible to decision-makers, I built a responsive Tableau dashboard consisting of three core analytical components and a set of user-focused features:

1. Executive KPIs

- Total underinsured exposure
- Number of high-impact counties
- Tier 1 & Tier 2 totals

2. Geospatial View

- County-level underinsurance tiers
- Color-coded underinsurance map highlighting regions requiring immediate attention

3. County Rankings & Model Diagnostics

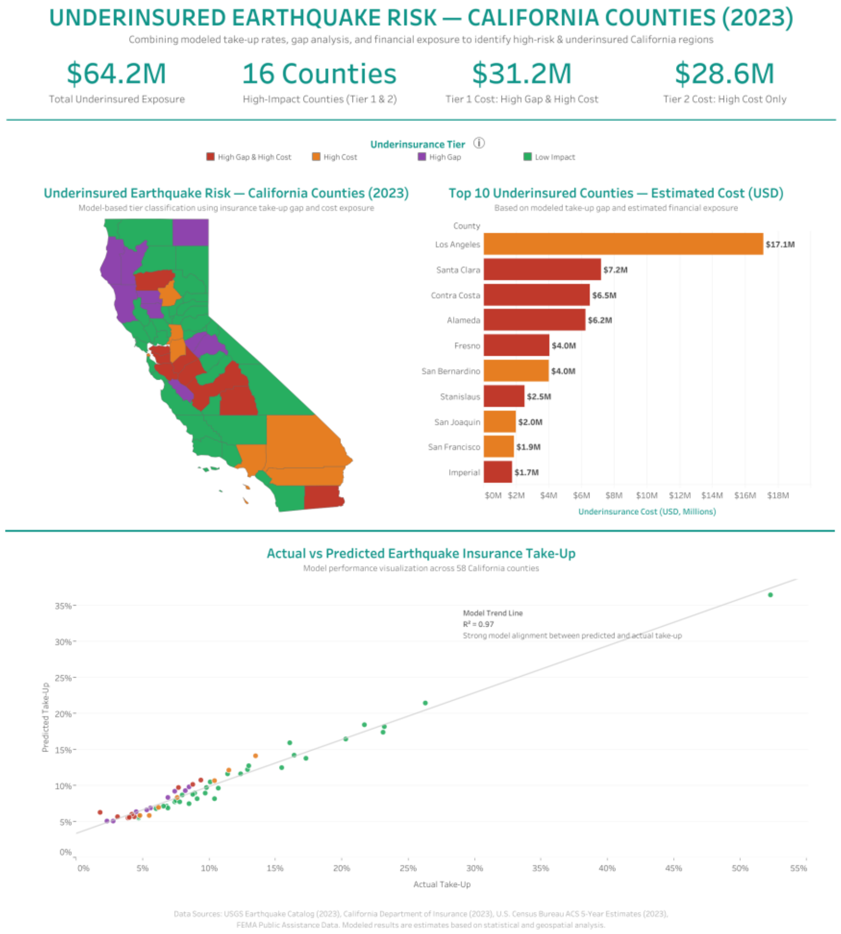
- Top 10 underinsured counties (USD)
- Scatterplot showing predicted vs. actual take-up
- Model trend line ($R^2 \approx 0.97$)

Dashboard Features

- Mobile-optimized layout
- Downloadable image, PDF, and PowerPoint
- Clear legend, coordinated color scheme
- Interactive tooltips with county-level model results

Dashboard Preview

Executive KPIs • Geospatial Map • County Rankings



Conclusion

This project demonstrates how integrating geospatial, seismic, socioeconomic, and insurance data can **reveal hidden underinsurance vulnerabilities across California.**

The predictive model reveals structural patterns in risk and coverage, while the underinsurance tiers offer a clear framework for identifying where financial exposure is most concentrated.

The Tableau dashboard presents these insights in an intuitive, decision-ready format — supporting insurance strategy, policy planning, and risk communication.

By integrating seismic exposure, vulnerability, and insurance behavior, this analysis **identifies where earthquake risk is most misaligned with actual protection.**

Limitations

1. Data resolution – County-level granularity may mask neighborhood- and parcel-level disparities
2. ACS timing – ACS socioeconomic variables are based on **5-year rolling averages**
3. CEA regions – Earthquake insurance data is reported by multi-county CEA regions (A–F), requiring regional-to-county mapping that may introduce **aggregation bias**
4. Policy granularity – CDI policy data does not differentiate **coverage limits**, deductibles, or policy types
5. Historical completeness – Historical seismic data is incomplete before modern instrumentation
6. Model fit – The predictive model explains major patterns (**$R^2 \approx 0.53$**) but leaves **meaningful variance unexplained**
7. Missing data – No building replacement cost or parcel-level exposure data were available

Future Enhancements

Data Enhancements

- Integrate **parcel-**, **ZIP-code-**, or **building-level data** for finer spatial resolution
- Add engineering-grade building attributes: **construction type**, **occupancy class**, and **replacement-cost estimates**

Modeling Enhancements

- Test additional modeling approaches (e.g., **CatBoost, LightGBM, GAMs**)
- Incorporate **uncertainty quantification**

Geographic Expansion

- Expand the framework to additional seismic states across the Western U.S.

Pipeline Enhancements

- Develop an automated **ETL pipeline** feeding a live, continuously updated Tableau dashboard

Policy Simulation

- Simulate **policy interventions** such as premium subsidies, take-up uplift scenarios, and targeted outreach

Thank You

Thank you for your time.

I'm happy to walk through the dashboard, modeling workflow, or EDA in more detail.

Anna Piterskaya

Data Science • Analytics • Geospatial Modeling

[GitHub Repository]

[Tableau Dashboard]